

float32: Store a value using 32 bits (binary digits)

(note 8 bits = 1 byte)

0 1 0 0 ... 0
32 in total

1024 bytes = 1 Kb

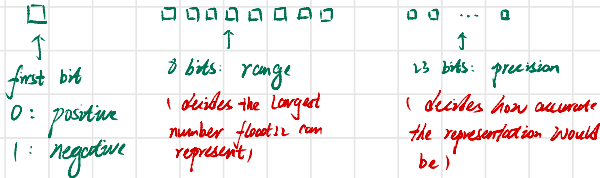
1024 Kb = 1 Mb

1024 Mb = 1 Gb

float16: Store using 16 bits

So 1 billion byte $\approx$ 1 Gb

How do we actually store a float number? (e.g. 1234.5678)
(using 32 bits)



(note: f32 or f64 may use different value of range/precision but the total bits add up to 32)

example 1: 1234.5678

Step 1: it is positive $\rightarrow$ first bit 0

Step 2: represent integer part: 1234 $\rightarrow$ 10011010010

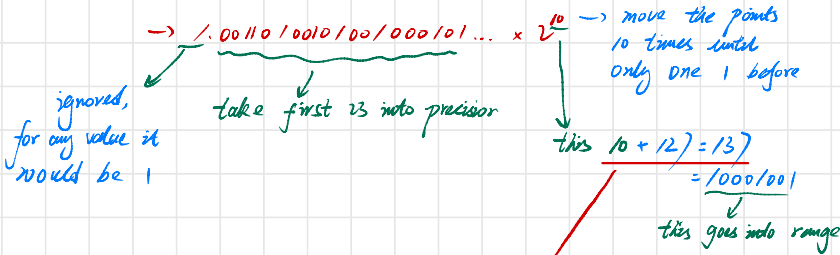
Step 3: represent fraction part: .5678 $\rightarrow$ 1001000101... (go on forever)

$$0.5678 \times 2 = 1.1356$$

$$0.1356 \times 2 = 0.2712$$

$$0.2712 \times 2 = 0.5424$$

So 1234.5678 $\rightarrow$ 10011010010.1001000101...



Q: why do we need to add 12 here?

A: if we store a super small float, e.g. 0.0000100234, the range would be 2^{-1} or something

$-1 + 12 = 11$ $\rightarrow$ this is still positive
so we save one bit for storing negative range